

In practice, the relationship of ΔV_{BE} to temperature is not perfectly linear, introducing measurement error. The accuracy is greatly improved by calibrating the sensor at a known temperature. Such calibration involves placing the chip or wafer in a thermal chamber and allowing time for temperature to equilibrate. The increased test time is expensive in a high-volume manufacturing environment. If thermal calibration is limited to 1 second, inaccuracies of about 0.5 °C can be achieved. Two-point calibration produces better results, but is impractically time-consuming.

13.3 Power Distribution

The power distribution subsystem of a chip consists of metal wires or planes on the chip, in the package, and on the printed circuit board. It also includes bypass capacitors to supply the instantaneous current requirements of the system. An ideal power distribution network has the following properties:

- Maintains a stable voltage with little noise
- Provides average and peak power demands
- Provides current return paths for signals
- Avoids wearout from electromigration and self-heating
- Consumes little chip area and wiring
- Easy to lay out

Real networks must balance these competing demands, meeting targets of noise and reliability as inexpensively as possible. The noise goal is typically $\pm 10\%$; for example, a system with nominal $V_{DD} = 1.0$ V may guarantee the actual supply remains within 0.9–1.1 V. Reliability goals demand enough vias and metal cross-sectional area to carry the supply current, as was discussed in Section 7.3.3. The two fundamental sources of power supply noise are IR drops and L di/dt noise.

Figure 13.6 plots the power consumption versus time for a microprocessor [Gauthier02]. The power varies on a number of time scales. While the processor is active, the power depends on the operations and data. It also spikes near the clock edges when the large clock loads switch. When the processor becomes idle, clock gating turns off the

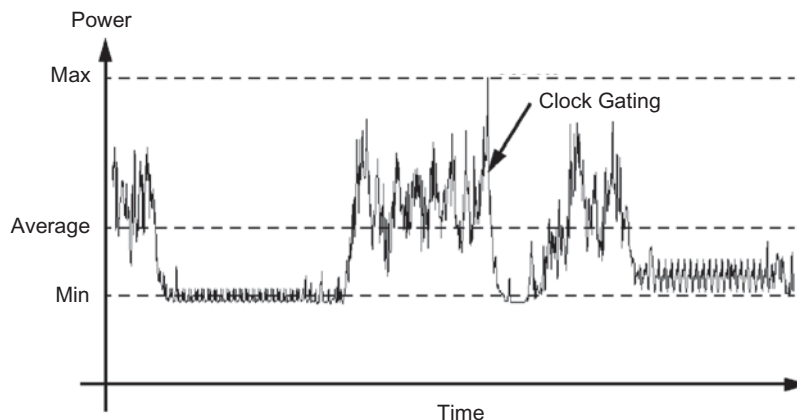


FIGURE 13.6 Time-dependent power consumption of microprocessor (Reprinted with permission of Sun Microsystems.)

clock to unused units, driving the power down significantly. As the supply voltage is nearly constant, the supply current I (also called I_{DD}) is proportional to the instantaneous power demand. As this current flows through the resistance R of the power distribution network, it causes a voltage droop proportional to IR . Moreover, as the changing current flows through the inductance of the printed circuit board and package, it also causes a voltage drop proportional to the rate of change: $L di/dt$.

This section begins by examining the physical design of a power distribution network. It then discusses IR drops and $L di/dt$ noise. The key to controlling noise from current spikes is to provide adequate bypass capacitance on and off the chip to provide low supply impedance at all frequencies. The power network is complicated enough that manual analysis is inadequate; instead, it typically must be modeled in a finite element simulation. The power network also provides return paths for current flowing in signal wires. The geometry of the network affects the inductance of on-chip signals. Some critical circuits such as phase-locked loops and analog blocks require a quiet supply for good performance. RC filters can reduce much of the supply noise. In sensitive circuits, noise carried through the substrate is also important.

13.3.1 On-Chip Power Distribution Network

The on-chip power distribution network consists of power and ground wires within the cells and more wires connecting the cells together. Most cells contain internal power and ground busses routed on metal1 or metal2. These wires are typically wider than minimum to provide lower resistance and better electromigration immunity. For example, the cells on the inside front cover use 8λ metal1 power/ground busses. These wires are normally connected between adjacent cells by abutment. Standard cell designs and datapaths both can use rows of cells sharing common power and ground lines.

In a small, low-power design, these rows can be strapped together with even wider vertical metal wires. Figure 13.7(a) shows an abstract diagram of this strapping. Figure 1.64 showed a standard cell design strapped with power on the left and ground on the right. In this example, the nMOS and pMOS transistors in adjacent rows are separated by a routing channel, so spacing between the wells is not a problem. In modern processes, the routing is typically done over the cell in upper-level metal. Therefore, the rows of cells can be packed more closely together and well spacing limits the packing density. Alternatively, every other row can be *mirrored* (flipped upside down) so that the wells of adjacent rows abut, as shown in Figure 13.7(b).

In a larger or high-power design, the resistance of the horizontal power and ground busses routed on thin lower-level metal will cause too much IR drop. Instead, the power should be delivered using a grid of metal on all layers. The top levels of metal are thickest and carry the bulk of the current, but a robust grid on all layers is important to bring the current down to the transistors. Where layers connect, multiple vias should be used to carry the high currents. As discussed in Section 6.3.4, the power and ground wires interdigitated with signal wires provide good return paths to minimize inductive effects. Systems with multiple voltage domains and/or power gating require particular attention to power network integrity [Kanno07].

The power grid extends across the entire chip or voltage domain. Ultimately, it must connect to the package through the I/O pads. When a pad ring is used, the connections are all near the periphery of the chip. Thus, the biggest IR drops occur near the center of the chip where the current flows through the longest wires and greatest resistance. C4 solder bumps distributed across the die are much better for power distribution because

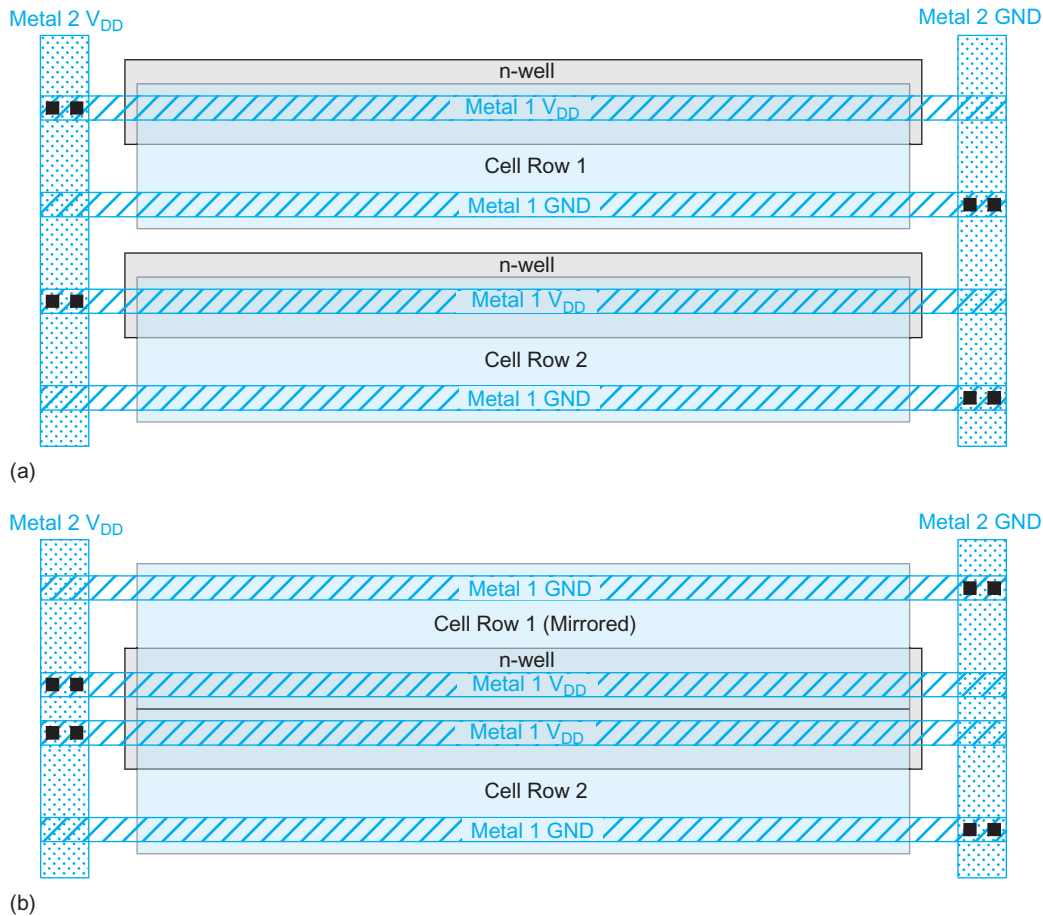


FIGURE 13.7 Power distribution for standard cell layout

they can deliver the current from the low-resistance power plane in the package directly to the area of the chip where the current is needed. Thus, less on-chip metal resources are needed for power distribution.

13.3.2 IR Drops

The resistance of the power supply network includes the resistance of the on-chip wires and vias, the resistance of the bond wires or solder bumps to the package, the resistance of the package planes or traces, and the resistance of the printed circuit board planes. Because the package and printed circuit board typically use copper that is much thicker and wider than on-chip wires, the on-chip network dominates the resistive drop.

IR drops arise from both average and instantaneous current requirements. The instantaneous current may be much larger than the average drop because current draw tends to locally spike near the clock edge when many registers and gates switch simultaneously. Bypass capacitance near the switching gates can supply much of this instantaneous current, so a well-bypassed power supply network only needs low enough resistance to deliver the average current demand, not necessarily the peak.

Example 13.3

Suppose a row of 64 repeaters share a common metal2 power bus like that shown in Figure 13.7(a). The bus is $320\text{ }\mu\text{m}$ long and $1\text{ }\mu\text{m}$ wide. The metal2 has a sheet resistance of $0.05\text{ }\Omega/\square$. If the repeaters drive 0.4 pF wire loads with 200 ps transition times, estimate the power supply droop seen by the repeater for a 1.8 V nominal supply.

SOLUTION: Each repeater draws a current of approximately

$$I = C \frac{\Delta V}{\Delta t} = (0.4\text{ pF}) \frac{1.8\text{ V}}{200\text{ ps}} = 3.6\text{ mA} \quad (13.7)$$

The power and ground busses each have a length of 320 squares and thus a resistance of $R = 16\text{ }\Omega$. The supply droop at the end of the wire caused by the 64 repeaters is $64 IR/2 = 1.85\text{ V}$, or more than V_{DD} , which is obviously impossible. Instead, as the power supply begins to droop, the repeaters deliver less current, reducing the droop, but increasing the transition time and delay. One way to alleviate this problem is to use a power grid so that each repeater obtains its current from its own vertical wire rather than sharing the single horizontal wire with all of the simultaneously switching neighbors. Figure 13.8 shows a simulation of one of the repeaters. It compares the two power bus layouts. When all the repeaters share a single power wire, the power supply droops by nearly 30% and the propagation delay is more than doubled. When each repeater has its own power wire so the supply noise is negligible, the output is crisper.

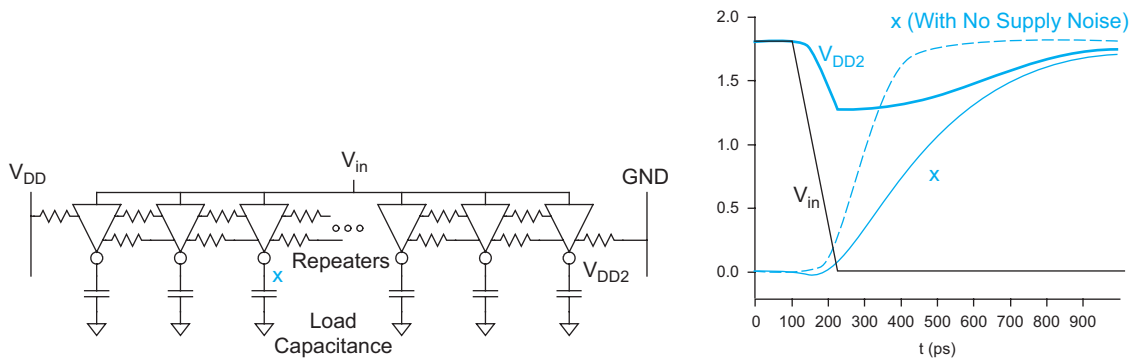


FIGURE 13.8 Power supply droop

13.3.3 L di/dt Noise

The inductance of the power supply is typically dominated by the inductance of the bond wires or C4 bumps connecting the die to the package. A typical bond wire has an inductance of about 1 nH/mm , while a C4 ball is on the order of 100 pH . Recall that the inductance of multiple inductors in parallel is reduced. Modern packages devote many (often 50% or more) of their pins or bumps to power and ground to minimize supply inductance. The two largest sources of current transients are switching I/O signals and changes between idle and active mode in the chip core.

Example 13.4

A 1 GHz chip transitions from idle to full power operation in a single cycle. The idle mode draws 20 A and the full power mode draws 60 A. If the power supply has 20 pH of series inductance, estimate the power supply noise caused by this transition if the chip has no internal bypass capacitance.

SOLUTION: The current transient is

$$\frac{\Delta I}{\Delta t} = \frac{(60 \text{ A} - 20 \text{ A})}{1 \text{ ns}} = 40 \text{ GA/s} \quad (13.8)$$

The inductive noise is $L \Delta I / \Delta t = 0.8 \text{ V}$. This is clearly unacceptable in a low-voltage system. Once again, the chip needs internal bypass capacitance to supply the instantaneous current, reducing the transient seen by the I/O pins.

$L di/dt$ noise is becoming enough of a problem that some high-power systems must resort to microarchitectural solutions that prevent the chip from transitioning between minimum and maximum power in a single cycle. For example, a pipeline may enter or exit idle mode one stage at a time rather than all at once to spread the current change over many cycles.

13.3.4 On-Chip Bypass Capacitance

As we have seen, chips need a substantial amount of capacitance between power and ground to provide the instantaneous current demands of the chip. This is called *bypass* or *decoupling* capacitance. The bypass capacitance is distributed across the chip so that a local spike in current can be supplied from nearby bypass capacitance rather than through the resistance of the overall power grid. It also greatly reduces the di/dt drawn from the package.

Example 13.5

How much bypass capacitance is needed to supply a sudden current spike of 40 A for 1 ns with no more than a 200 mV supply droop?

SOLUTION: We solve

$$I = C \frac{\Delta V}{\Delta t} \Rightarrow C = \frac{(40 \text{ A})(1 \text{ ns})}{0.2 \text{ V}} = 200 \text{ nF} \quad (13.9)$$

Fortunately, the inherent gate capacitance of quiescent transistors provides a significant amount of *symbiotic* bypass capacitance [Dally98]. For example, Figure 13.9 shows one inverter driving another. The gate-to-source capacitances of the load inverter are shown explicitly. When $A = 1$ and $B = 0$, $M1$ is ON, charging up C_{gs4} . Similarly, when $A = 0$ and $B = 1$, $M2$ is ON, charging up C_{gs3} . The charged capacitor stores energy that can be released to supply sudden current demands. At any given time, approximately half of the gate capacitance of any quiescent circuit behaves as symbiotic bypass capacitance. Moreover, because only a small fraction of the gates are likely to be switching at any given time, nearly half of the entire gate capacitance on the chip will serve as bypass capacitance.

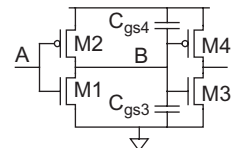


FIGURE 13.9
Symbiotic bypass capacitance

Example 13.6

Estimate the symbiotic bypass capacitance per square millimeter for a chip with feature size f if gate capacitance is $1 \text{ fF}/\mu\text{m}$ of transistor width and transistor gates occupy 9% of chip area.

SOLUTION: The capacitance density of a $1 \mu\text{m}$ wide transistor of length f is $1/f \text{ fF}/\mu\text{m}^2$. At 9% utilization, this corresponds to $0.09/f \text{ nF}/\text{mm}^2$. Half of that, or $0.045/f \text{ nF}/\text{mm}^2$, serves as symbiotic bypass capacitance on average. In an $f = 65 \text{ nm}$ process, this means the symbiotic bypass capacitance is approximately $0.7 \text{ nF}/\text{mm}^2$.

In most low- and medium-power chips, this symbiotic capacitance provides adequate bypassing to filter instantaneous IR drops and $L \text{ di}/\text{dt}$ noise. In high-power chips, additional explicit capacitance is necessary. For example, the Sun Niagara2 processor added 700 nF of on-chip decoupling capacitance [Nawathe08]. The only dielectric available in a standard CMOS process to build compact high-capacitance structures is gate oxide, so the extra bypass capacitance is commonly built with an nMOS transistor with the gate tied to V_{DD} and the source and drain tied to GND. Decoupling capacitor layout should maximize the capacitance per unit area. [Meng08] describes bypass capacitor layout techniques.

In some nanometer processes, gate leakage is significant for thin-oxide transistors. Thicker-oxide transistors may be preferable to save leakage at the cost of lower capacitance density. Sun used thick-oxide transistors in the Rock processor with a 20% loss in capacitance density [Konstadinidis09].

13.3.5 Power Network Modeling

Figure 13.10 shows a lumped model of the power distribution network for a system including the voltage regulator, the printed circuit board planes, the package, and the chip. The network also includes bypass capacitors near the voltage regulator, near the chip package, possibly inside the chip package, and definitely on chip. The external capacitors are modeled as an ideal capacitor with an effective series resistance (ESR) and effective series inductance (ESL) representing the parasitics of the capacitor package. Larger capacitors have bigger effective series inductances.

The voltage regulator seeks to produce a constant output voltage independent of the load current. It is modeled as an ideal voltage source in series with a small resistance and the inductance of its pins. Near the regulator is a large bulk capacitor (typically electrolytic or tantalum). Power and ground planes on the printed circuit board carry the supply

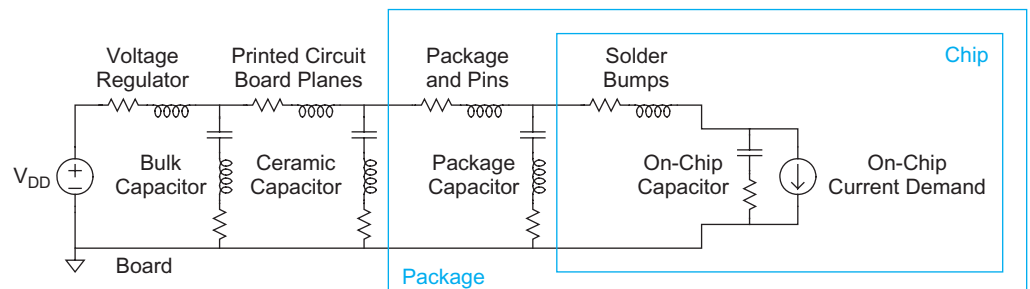


FIGURE 13.10 Power distribution system model

current to the package, contributing some resistance and inductance. Typically, the board designer places several small ceramic capacitors near the package. The package and its pins again contribute resistance and inductance. High-frequency packages often contain small capacitors inside the package for further decoupling. Finally, the chip connects to the package through solder bumps or bond wires with additional resistance and inductance. The dynamic and static current demands of the chip are modeled as a variable current source with a waveform that might resemble Figure 13.6. The on-chip bypass capacitance consists of the symbiotic capacitance and possibly some explicit decoupling capacitance. It typically has negligible inductance because it is located so close to the switching loads.

As one moves from the chip toward the voltage regulator, each capacitor typically increases by about an order of magnitude. However, each series inductance increases by a similar amount. [Budnik06] illustrates a representative power delivery network for a high-performance 90 nm microprocessor. The capacitance is on the order of 1 μF on the die, 10's of μF in the package, and 100's of μF on the board, and 1 mF at the voltage regulator. The inductance is on the order of 1 pH between the die and package, 10 pH between the package and board, and 100 pH along the board to the voltage regulator. The resistance is a fraction of an m Ω at each link.

13.3.5.1 Power Supply Impedance A good power distribution network should offer a low impedance at all frequencies of interest so that the supply voltage remains steady independent of the changing chip current demands. If the system draws P watts of power and the maximum allowable power supply ripple is $r \times V_{DD}$ (e.g., $r = 0.1$ for 10% supply noise), then the supply impedance must be less than

$$Z = \frac{rV_{DD}^2}{P} \quad (13.10)$$

This relationship shows that required supply impedance is dropping quadratically with voltage scaling. It is also dropping as power consumption increases. This impedance requirement has driven the adoption of improved packages and flip-chip bonding with solder bumps instead of bond wires. It means chips need to use more metal and on-chip bypass capacitance. For example, a 1.0 V system dissipating 100 W of power draws 100 A. To keep supply noise down to 10% of V_{DD} , the power supply impedance must be 1 m Ω .

If the system had no bypass capacitance, the distribution network would consist of only the resistance and inductance, so it would have an impedance of $Z = R + j\omega L$. This impedance increases with frequency ω and becomes unacceptably high for most systems by about 1 MHz.

The bypass capacitors in parallel with the supply provide an alternative low-impedance path at higher frequencies. An ideal capacitor has impedance that decreases with frequency as $Z = 1/j\omega C$. Unfortunately, the effective series inductance of the capacitors limits the useful frequency range of the real capacitor. The impedance of a capacitor C with effective series resistance R and inductance L is

$$Z = \frac{1}{j\omega C} + R + j\omega L \quad (13.11)$$

This impedance has a minimum of $Z = R$ at the self-resonant frequency of

$$f_{\text{resonant}} = \frac{\omega_{\text{resonant}}}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \quad (13.12)$$

Figure 13.11 plots the magnitude of the impedance of a $1\ \mu\text{F}$ capacitor with $0.25\ \text{nH}$ of series inductance and $0.03\ \Omega$ of series resistance. The capacitor has low impedance near its resonant frequency of $10\ \text{MHz}$, but higher impedance elsewhere.

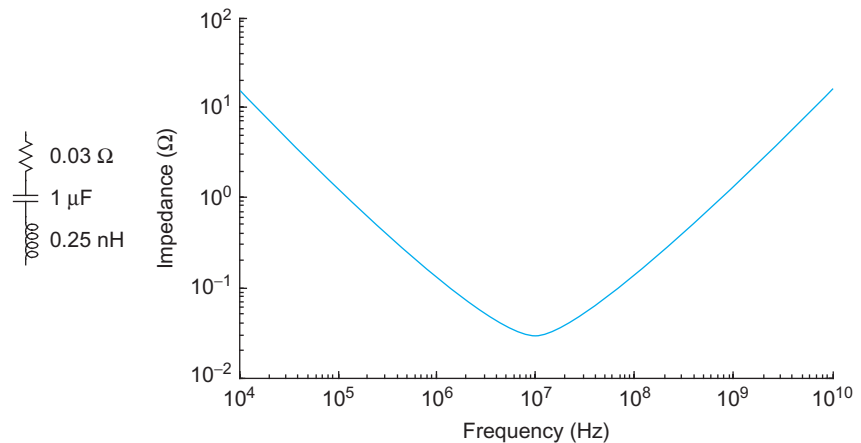


FIGURE 13.11 Impedance of bypass capacitor

Larger capacitors tend to have higher effective series inductances and therefore have lower self-resonant frequencies beyond which they are not useful. Thus, the system uses many capacitors of different sizes to provide low impedance over all the frequencies of interest. Also, capacitors closer to the chip are more useful at high frequencies because they have less inductance in the board and package between them and the chip. The bulk and ceramic capacitors are most effective over the $1\text{--}10\ \text{MHz}$ range. Capacitors in the package tend to be useful in the $10\text{--}200\ \text{MHz}$ range. Above a few hundred MHz, the inductance of the solder bumps or bond wires renders all but the on-chip decoupling capacitors ineffective.

Figure 13.12 shows the simulated impedance of the Pentium 4 power distribution network as a function of frequency, illustrating the resonances caused by the package, socket, board, and regulator [Xu08]. Note the large increase in impedance near $100\ \text{MHz}$ caused by the package.

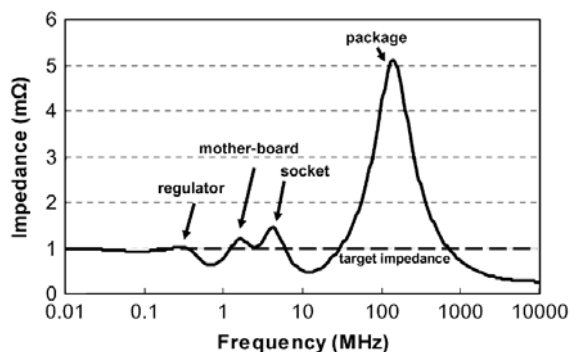


FIGURE 13.12 Pentium 4 simulated power supply impedance (© 2008 IEEE.)

13.3.5.2 Power Supply Step Response Another way to think about the need for nearby bypass capacitance is to imagine a sudden step in current on the chip. Some round-trip propagation delay must occur before the spike reaches the power supply, the supply adjusts the current it is delivering, and that current returns to the chip. A lower bound on this delay is the speed of light. Therefore, when a gate switches, the voltage regulator will not know about the event until sometime after the transition has completed. In the meantime, the charge must be drawn from the bypass capacitors. This results in a series of droops as each capacitor becomes depleted before the next one kicks in. Yet another perspective is to remember that an inductor does not like to change its current instantaneously. Thus, larger inductors introduce a longer lag.

Figure 13.13 shows the simulated response to an abrupt increase in current demand on the Pentium 4 illustrating a sequence of three droops characteristic of power distribution networks [Wong06]. Before the step, the voltage regulator delivers some amount of current sufficient to meet the average needs of the chip. When the current demand suddenly increases, the extra charge is initially drawn out of the on-chip bypass capacitors. As these capacitors discharge, the supply voltage drops precipitously. This is called the *first droop*. Soon the current through the solder bumps increases to recharge the on-chip capacitors. The delay depends on the inductance of the bumps. Moreover, this inductance may cause the supply voltage to overshoot and oscillate. Meanwhile, the capacitors in the package supplying this current start to discharge and the voltage droops again. This *second droop* occurs on a longer time scale determined by the package capacitance. Eventually, the current through the package pins and socket increases to begin recharging the package capacitors. Meanwhile, the capacitors on the printed circuit board discharge, leading to a *third droop* before the voltage regulator catches up with the increased current demand. The second and third droops are minimized by providing an adequate number of high-quality, low ESL capacitors at each stage in the power distribution network [Smith99].

Designers typically assume that adding on-chip bypass capacitance to reduce supply droop improves operating frequency. While more capacitance certainly does reduce the droop, the frequency does not necessarily improve. In a striking experiment, [Wong06] fabricated several wafers of Pentium 4 processors with and without decoupling capacitors. Without capacitors, the first droop increased by 8% of V_{DD} , but the operating frequency only slowed by 1%. The anomaly was explained by showing that under certain conditions the noise modulates the clock period in a way that tracks the critical path delay.

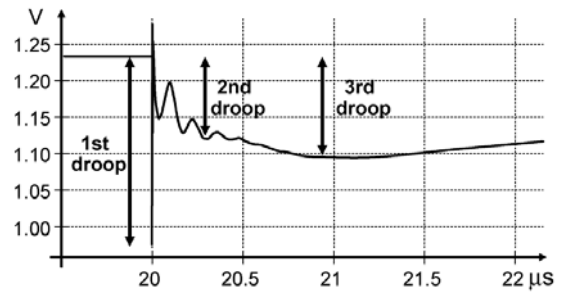


FIGURE 13.13 Pentium 4 simulated voltage droops
(© 2006 IEEE.)

13.3.5.3 Distributed Power Supply Models The model presented so far is a lumped approximation that is convenient for analysis and facilitates gaining intuition about chip behavior. Chip designers also are concerned about the variation in supply voltage across the chip. This requires a distributed model, which we can approximate with a mesh of small elements as shown in Figure 13.14. The mesh represents the resistance and inductance of the

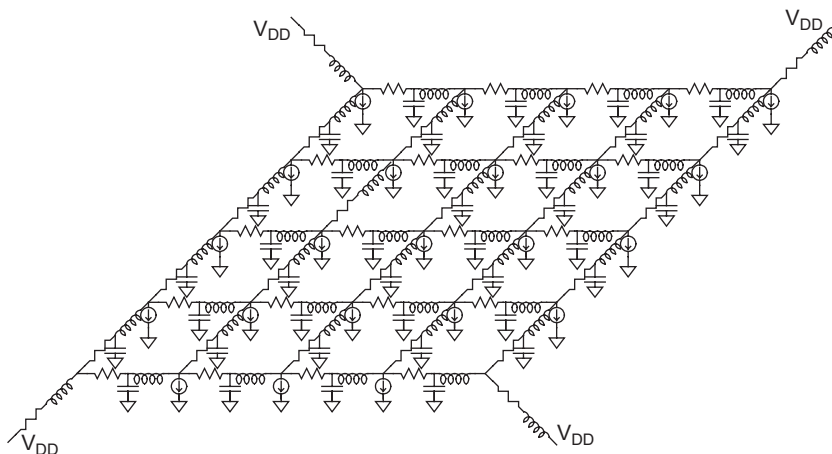


FIGURE 13.14 Impedance of bypass capacitor

on-chip power supply grid. Symbiotic or explicit decoupling capacitors are distributed across the chip. At each node, a current source represents the local current demand of the circuitry. The solder bumps or bond wires to the package are modeled with additional resistance and inductance. In this model, the package is treated as a perfect V_{DD} connected to the corners of the grid. In a more complex model, you also could add the distributed resistance, inductance, and bypass capacitance of the package itself.

For high-power chips, the designer can extract a mesh model as a SPICE netlist based on the power grid wiring and the amount of local decoupling capacitance. Different current waveforms can be applied at different nodes; for example, the current signatures of synthesized logic, SRAM, repeater banks, and domino logic are all quite different. The full-chip power grid simulation often takes many days to run and results in a map of voltage vs. time for the current pattern applied. Figure 7.2 shows a snapshot of the voltage droop on the Itanium 2 microprocessor. The droop was greatest in the integer execution unit, where several power-hungry domino adders all contribute to the IR drop.

13.3.6 Power Supply Filtering

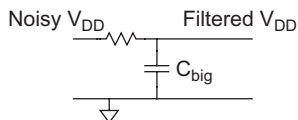


FIGURE 13.15
Power supply filter

Certain structures such as the phase-locked loop (PLL), clock buffers, and analog circuits are particularly sensitive to power supply noise. For example, supply noise on the clock buffers can directly increase clock jitter. Figure 13.15 shows an RC power supply filter circuit that eliminates the high-frequency noise on the local supply. The local filtered power supply is typically connected to the power grid through a single wire or solder bump. The resistance of this wire must be low enough to carry the current demand of the local circuitry without excessive IR drop, yet low enough to produce an RC time constant that will filter noise at frequencies of interest. Typically, this requires a huge filter capacitor as well, making power supply filtering expensive in terms of chip area.

For example, the Pentium 4 uses a power supply filter on the clock buffers to reduce clock jitter [Kurd01]. The filter attenuates typical supply noise from 10 to 2% of V_{DD} using a pMOS transistor as the resistor. It has an RC time constant of 2.5 ns with an IR drop of 70 mV.

13.3.7 Charge Pumps

Many circuits require a positive voltage exceeding V_{DD} or a negative voltage. For example, a Flash memory may require 20 V to erase floating-gate transistors. Reverse body bias techniques need a negative voltage. Extra external voltage regulators add to the system cost. If the current requirements are not too high, these voltages can be generated on-chip using a *charge pump*.

Figure 13.16 shows a Dickson charge pump [Dickson76]. The pump uses two non-overlapping clock phases. Initially, node V_1 is charged up to $V_{DD} - V_t$ through $N1$. When ϕ_1 rises, the capacitor drives V_1 up. When $V_1 - V_2 > V_t$, $N2$ turns ON and begins charging V_2 toward $2(V_{DD} - V_t)$. When ϕ_1 falls, the capacitor drags node V_1 back down. $N2$ turns OFF, leaving V_2 at the elevated voltage. Next, ϕ_2 rises, pushing up V_2 and V_3 toward $3(V_{DD} - V_t)$. The pumping continues down the line. With enough stages, V_{out} can be driven arbitrary high, subject to limitations such as breakdown. The pumping capacitors C can be constructed out of nMOS transistors with their source and drain connected to the clock and the gate tied to the node being pumped. Larger capacitors pumped at higher frequency f increase the available output current. If each of the pumped nodes has a stray capac-

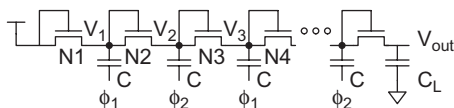


FIGURE 13.16 Dickson charge pump

itance C_s (such as the gate and diffusion capacitance of the transistors to the right and left) then the output voltage is approximately

$$V_{\text{out}} = N \left[\frac{CV_{DD} - \frac{I_{\text{out}}}{f}}{C + C_s} - V_t \right] \quad (13.13)$$

A large load capacitor C_L helps smooth out the ripple on V_{out} .

Figure 13.17 shows a charge pump for negative voltages. The pump operates in a similar fashion, but the pMOS transistors pull the voltage down on the falling transition of the clock. The pMOS bodies can be tied to GND to reduce the body effect.

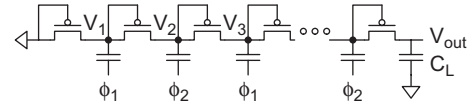


FIGURE 13.17 Negative charge pump

13.3.8 Substrate Noise

The body terminal of a bulk CMOS transistor is connected to the substrate or well. The p-type substrate for an nMOS transistor is normally connected to GND and the n-well for a pMOS transistor is normally connected to V_{DD} . The connection is made through a relatively high-resistance substrate or well contact. Current flow in the substrate causes noise on the body terminal. This current may come from capacitive coupling through the reverse-biased source/drain to substrate diodes or from impact ionization as current flows through an ON transistor. The substrate noise modulates threshold voltages by means of the body effect.

Substrate noise is also a problem for *mixed-signal* designs where separate power supplies are used for noisy digital circuits and quiet analog circuits. The large number of rapidly switching digital circuits creates noise on the digital ground that propagates to the sensitive analog circuitry via the common substrate.

The substrate and well should use plenty of contacts to guarantee a low-resistance path to the power network. Guard rings, described in Section 7.3.6, provide some protection against noise caused by currents in the nearby substrate. Analog circuits should be physically separated from digital circuits and protected by guard rings connected to a quiet analog supply. Twin-tub or triple-well processes and SOI also experience much less substrate coupling because transistors are isolated in their own wells.

Modeling and analyzing substrate noise is beyond the scope of this book. See [Donnay03] for extensive coverage of the subject.

13.3.9 Energy Scavenging

Energy sources are a chronic challenge for portable systems. Most systems use batteries, which eventually require replacement. This ranges from annoying (remembering to change the battery in your fire alarm each year) to downright difficult (changing the battery in an implanted pacemaker). *Energy scavenging* is an emerging field with tremendous promise for ultra-low power systems. The idea is to extract enough energy from the environment to operate the device. The technique is particularly attractive when combined with subthreshold circuits operating at microwatt or nanowatt average power levels. The power demand typically varies with time, so the energy may be stored in a capacitor or *microbattery* until it is needed.

Micropower generators can take advantage of many sources of energy. *Solar cells* are the best known [Guilar09]; solar calculators are among the oldest and best-known

applications of energy scavenging. *Thermoelectric* microgenerators use thermocouples to produce a voltage proportional to the temperature difference across the elements [Lhermet08]. *Piezoelectric* microgenerators convert mechanical vibration into electricity [Le06, Ramadass10]. Radio-frequency identification (RFID) tags use a coil to collect RF energy radiated by the reader, then broadcast their ID back. Power output for these sources depends on the amount of energy available for scavenging and the size of the generator, but tens to hundreds of microwatts per square centimeter are commonly achieved.

Microbattery manufacturing is also evolving. Microbatteries are made from layers of thin films that can be deposited on top of an integrated circuit after the standard steps are completed. A 10- μm thick lithium-based battery presently achieves an energy density of 100 $\mu\text{W}\cdot\text{hr}/\text{cm}^2$ [Lhermet08].

13.4 Clocks

Synchronous systems use a clock to distinguish one step in a computation from the previous or next step. Ideally, this clock should arrive at all clocked elements in the system simultaneously so that the system shares a common time reference. These elements include latches and flip-flops, memories, and dynamic gates. In practice, the arrival time differs somewhat from one point to another; this difference is called *clock skew*. The central challenge in clock system design is to deliver the clock to all the clocked elements on the chip while finding an acceptable compromise among skew, power consumption, metal resource usage, and design effort.

13.4.1 Definitions

A system is designed to use one or more *logical clocks*. The logical clocks are idealized signals with no skew used by the logic designer when describing the system with a hardware description language. For example, a system with flip-flops requires a single logical clock, usually called *clk*. A system using two-phase transparent latches requires two logical clocks ϕ_1 and ϕ_2 (or *ph1* and *ph2* in a hardware description language). Unfortunately, mismatched clock network paths and processing and environmental variations make it impossible for all clocks to arrive at their ideal times, so the designer must settle for actually receiving a multitude of skewed *physical clocks*.

Distributing a single clock across the entire chip in a low-skew fashion is challenging. Distributing more than one is nearly impossible. Therefore, most systems distribute a single *global clock* even though they may need multiple logical clocks. *Local clock gates* located near the clocked elements produce the physical clocks and drive them to the elements over short wires. Examples of clock gates include buffers, AND gates to stop the clock to unused units, inverters to produce complementary clocks, and pulse generators for pulsed latches.

The term *clock skew* has been used informally in many ways. We define skew as the *difference between the nominal and actual interarrival time of a pair of physical clocks*. For example, Figure 13.18(a) shows a system with two flip-flops. Both should receive the logical clock *clk* with zero interarrival time, but they actually receive physical clocks clk_1 and clk_2 . Because of differences in the delay of the clock distribution wires and the local clock buffers, clk_1 arrives 25 ps before clk_2 . Therefore, we say the clock skew is 25 ps. Figure 13.18(b) shows a system with three transparent latches. The latches use complementary logical clocks ϕ_1 and ϕ_2 with a nominal interarrival time of $T_c/2$ between rising edges.